

Unsupervised Phenotyping of Cardiac Risk: A Clustering and Bayesian Network Approach to the UCI Heart Disease Dataset

Omar Degui
Matr. Num. 888573
MSc Artificial Intelligence
for Science and Technology
Email: o.degui@campus.unimib.it

Lorenzo Luchesini
Matr. Num. 881345
MSc Artificial Intelligence
for Science and Technology
Email: l.luchesini@campus.unimib.it

Matteo Rigoni
Matr. Num. 883831
MSc Artificial Intelligence
for Science and Technology
Email: m.rigoni3@campus.unimib.it

Abstract—Can distinct cardiac risk phenotypes be recovered from routine clinical measurements without any diagnostic label? This report investigates the question through a fully unsupervised pipeline applied to the UCI Heart Disease dataset (920 patients, 15 raw attributes). Following a rigorous preprocessing stage—missing-value imputation, one-hot encoding, standardisation, and Isolation Forest outlier removal—the dimensionality is reduced with Principal Component Analysis (PCA) retaining 90% of the variance, mitigating the curse of dimensionality that afflicts distance-based methods. Three complementary clustering paradigms are then compared: centroid-based K-means++, hierarchical Agglomerative (Ward) clustering, and density-based DBSCAN. Cluster quality is assessed exclusively through internal validation, principally the Silhouette Coefficient cross-checked with the Davies–Bouldin index. Finally, a Bayesian Network is learned by Hill-Climb structure search scored with the Bayesian Information Criterion (BIC), yielding an interpretable directed acyclic graph in which the discovered cluster label emerges as a central latent risk stratifier. The study emphasises convergence behaviour, initialisation stability, noise robustness, the role of feature scaling, and the computational scalability of each technique toward real-time deployment.

Index Terms—unsupervised learning, clustering, K-means, DBSCAN, hierarchical clustering, PCA, Bayesian networks, internal validation, curse of dimensionality.

I. INTRODUCTION

Cardiovascular disease remains the leading cause of mortality worldwide, and the early identification of at-risk subpopulations is of considerable clinical value. Supervised classifiers can predict a diagnostic label when one is available; however, in many practical settings labels are noisy, incomplete, or absent altogether. This motivates an *unsupervised* formulation of the problem: given only routine clinical measurements, can natural groupings of patients be discovered, and do these groupings correspond to interpretable cardiac phenotypes?

The present work addresses this question using the UCI Heart Disease dataset [1], comprising $N = 920$ patient records described by demographic, haemodynamic, and electrocardiographic attributes. The analysis pursues two objectives. First, three structurally distinct clustering algorithms are applied and

rigorously evaluated to partition the patient population. Second, a probabilistic graphical model—a Bayesian Network—is learned over the clinical variables augmented with the discovered cluster label, in order to expose the conditional dependencies that characterise each phenotype.

The remainder of the paper is organised as follows. Section II describes the dataset and exploratory analysis. Section III details the preprocessing pipeline, with particular attention to feature scaling and outlier handling. Section IV discusses dimensionality reduction as a remedy for the curse of dimensionality. Section V presents the three clustering techniques and the rationale for their selection. Section VI defines the notion of optimal clustering through internal validation and reports the results. Section VII analyses convergence, local minima, and stability. Section VIII develops the Bayesian Network and its information-criterion-based structure search. Section IX discusses computational complexity and scalability, and Section X concludes.

II. DATA AND EXPLORATORY ANALYSIS

The dataset contains 920 objects and 15 attributes [1]. Continuous clinical measurements comprise age (*age*), resting blood pressure (*trestbps*), serum cholesterol (*chol*), maximum heart rate achieved (*thalch*), and ST-depression (*oldpeak*). The remaining variables are categorical or boolean, including chest-pain type (*cp*), fasting blood sugar (*fbs*), resting electrocardiographic results (*restecg*), exercise-induced angina (*exang*), and the ST-segment slope (*slope*).

Exploratory data analysis focused on schema inspection, cardinality of each attribute, and—critically—the per-column missingness ratio, which drives the cleaning strategy. As shown in Figure 1, only two attributes, the number of major vessels (*ca*) and the thalassemia indicator (*thal*), exceed the 50% missingness threshold and are therefore discarded as non-recoverable. Figure 2 reports the distributions of the five

continuous features, which are visibly skewed and span heterogeneous ranges; this confirms that median-based imputation and subsequent standardisation are the appropriate choices.

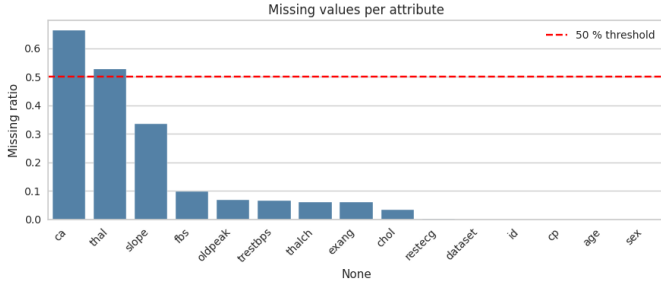


Fig. 1. Missing-value ratio per attribute. The dashed line marks the 50% threshold above which columns (ca, thal) are dropped as non-recoverable.

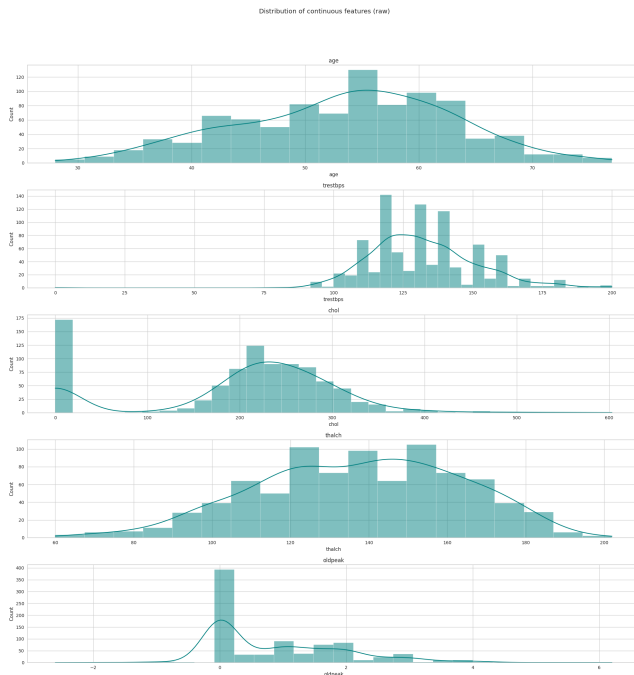


Fig. 2. Distribution of the five continuous clinical features prior to preprocessing. The differing supports and variances motivate subsequent standardisation.

III. PREPROCESSING

A. Cleaning, Encoding, and Imputation

Two non-informative columns, the record identifier (id) and the acquisition site (dataset), are removed since they carry no clinical signal. After dropping the high-missing attributes, remaining gaps are filled by *median* imputation for numeric features and *most-frequent* (mode) imputation for categorical features, a robust choice that resists the influence of skew and outliers.

Nominal and boolean attributes are transformed via one-hot encoding (drop_first=True to avoid collinearity), yielding an exclusively numerical feature matrix. This design choice deliberately avoids assigning arbitrary ordinal integers

to unordered categories, which would inject a false notion of distance into the subsequent Euclidean geometry.

B. Feature Scaling and Distance Parity

Distance-based clustering is acutely sensitive to feature scale. When attributes are expressed on vastly different ranges, the Euclidean distance is dominated by the highest-variance dimension, and the partition effectively degenerates into a univariate split on that axis. In the raw data the variance of serum cholesterol (cholesterol) is roughly two orders of magnitude larger than that of age; unscaled clustering therefore slices the cholesterol axis almost exclusively.

To guarantee feature parity, the numeric columns are standardised with `StandardScaler` to zero mean and unit variance, so that every dimension contributes comparably to the pairwise distance. Figure 3 contrasts the numeric features before and after this transformation, showing that standardisation equalises their spread. The consequences of omitting the step are quantified experimentally in Section VI (Table III): the apparently high silhouette obtained on the unscaled matrix is an artefact of the cholesterol split and collapses once standardisation restores parity.



Fig. 3. Continuous features before and after `StandardScaler`. Standardisation centres and rescales every attribute to comparable spread, ensuring no single variable dominates the Euclidean metric.

C. Outlier Detection with Isolation Forest

Anomalous records are identified with an Isolation Forest [2] configured with a contamination rate of 0.05. The algorithm isolates points via random recursive partitioning; anomalies, being few and distinct, require fewer splits to isolate and are flagged efficiently. Applied to the standardised matrix, the model flagged 46 observations (approximately 5% of the data), which were removed prior to clustering. An

aligned pre-dummy copy of the cleaned data was retained for the Bayesian Network stage of Section VIII. The resulting clean feature matrix serves as the input to dimensionality reduction. The Euclidean proximity matrix in Figure 4 already exhibits a faint block structure, providing preliminary visual evidence that separable groups are present.

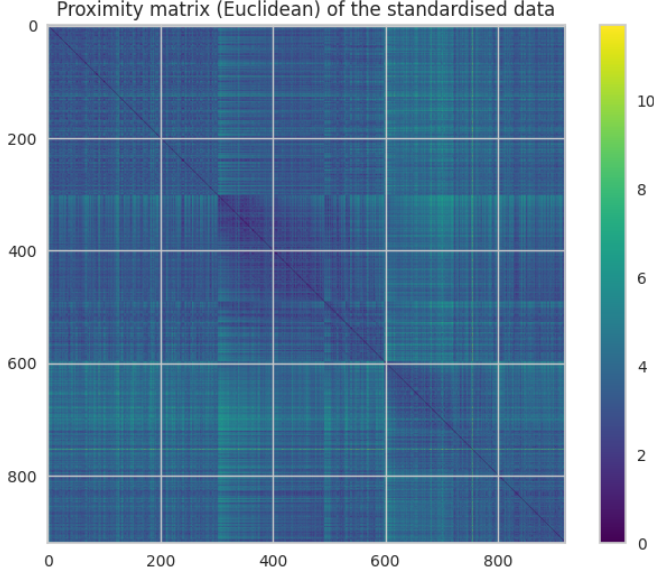


Fig. 4. Euclidean proximity matrix of the standardised data. Emerging block structure provides preliminary visual evidence of separable groups.

IV. DIMENSIONALITY REDUCTION

One-hot encoding inflates the feature space, and in high dimensions Euclidean distances *concentrate*: the contrast between the nearest and farthest neighbours of a point vanishes, undermining any distance-based notion of a cluster. This *curse of dimensionality* is mitigated by projecting the data onto a compact subspace with Principal Component Analysis (PCA).

The number of components is selected to retain 90% of the total variance, which reduces the encoded 19-dimensional space to 10 principal components. As shown in Figure 5, ten components suffice to cross the 90% cumulative-variance threshold, so the remaining directions are discarded as noise. Clustering is performed in this reduced space, where distances are better behaved and noise directions are suppressed. Separately, t-distributed Stochastic Neighbour Embedding (t-SNE, perplexity = 20) is employed *only* as a two-dimensional visual aid for inspecting cluster separation (Figure 6); it is never used as the input geometry for clustering, since its non-parametric, non-convex objective would confound the quantitative comparison of algorithms.

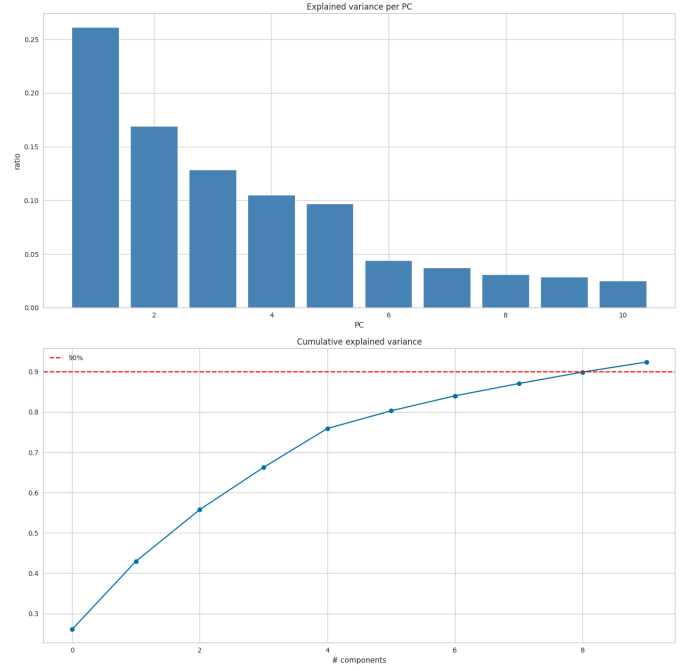


Fig. 5. Per-component (top) and cumulative (bottom) explained variance. Ten principal components suffice to cross the 90% threshold marked by the dashed line.

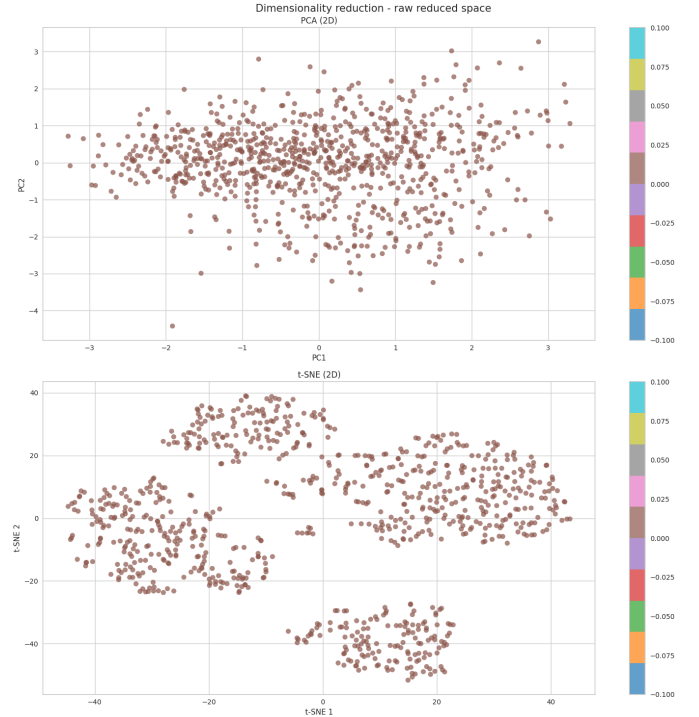


Fig. 6. PCA (2D) and t-SNE (2D) projections of the reduced feature space prior to clustering, used as a visual diagnostic of latent structure.

V. CLUSTERING METHODOLOGY

A. Rationale for Algorithm Selection

Three algorithms were deliberately chosen to represent three *structurally distinct* families, so that agreement between them

TABLE I
SELECTED CLUSTERING ALGORITHMS AND THEIR INDUCTIVE BIASES.

Algorithm	Family	Parameter selection
K-means++	Centroid	Elbow (distortion) jointly with silhouette
Agglomerative (Ward)	Hierarchical	Silhouette on the dendrogram cut
DBSCAN	Density	k -distance knee, refined by silhouette grid

constitutes stronger evidence of genuine structure than any single method could provide. K-means++ is a centroid-based partitioner that assumes convex, roughly isotropic clusters and requires the number of clusters k in advance. Agglomerative clustering with Ward linkage is a hierarchical method that reveals the nested merge structure of the data and imposes no distributional assumption beyond a variance-minimising merge criterion. DBSCAN is a density-based method that requires no prior k , recovers arbitrarily shaped clusters, and explicitly labels low-density points as noise. Their complementary inductive biases are summarised in Table I.

B. K-means++

K-means clustering has previously been applied to heart-disease data in an analogous unsupervised setting [3], motivating its inclusion here. It is initialised with the `k-means++` seeding scheme and $n_{\text{init}} = 10$ restarts, retaining the solution of lowest within-cluster sum of squares (SSE). The number of clusters is chosen by inspecting the distortion elbow and the silhouette curve jointly over $k \in [2, 10]$. As Figure 7 and Figure 8 show, the elbow and the silhouette peak agree on $k = 3$, which is adopted as the primary criterion. The resulting partition is visualised in Figure 9.

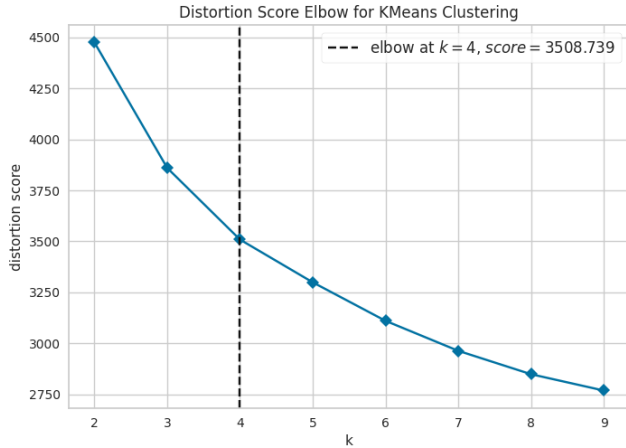


Fig. 7. Model selection for K-means++: distortion elbow as a function of k .

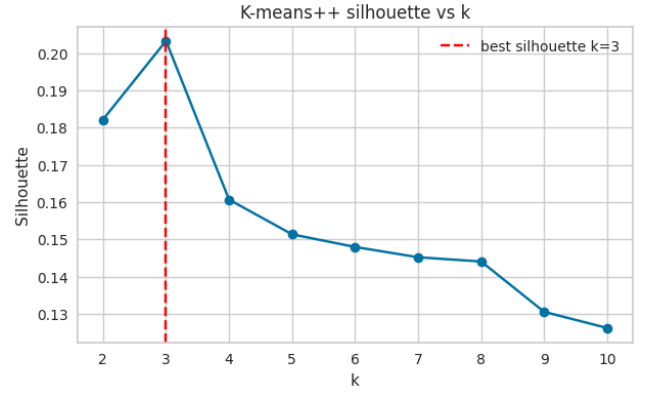


Fig. 8. Model selection for K-means++: silhouette score as a function of k . The silhouette peak identifies the retained value of k .

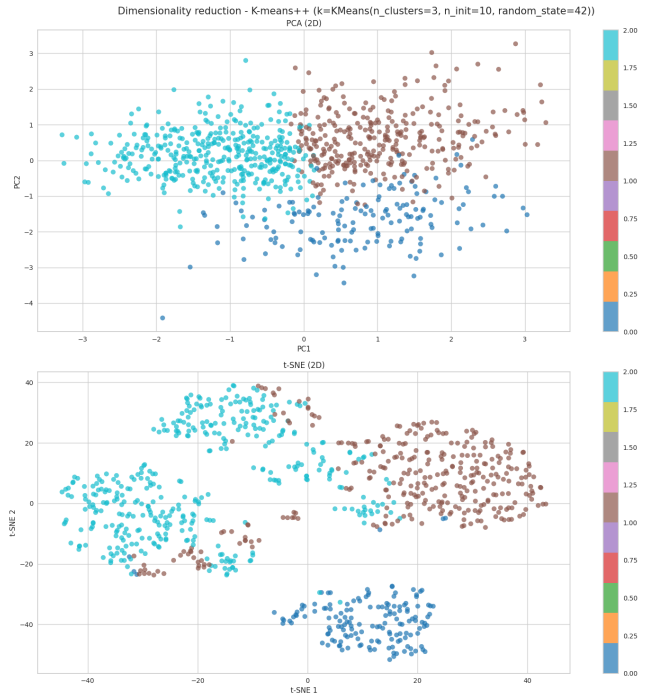


Fig. 9. K-means++ partition visualised in the PCA and t-SNE projections.

C. Agglomerative (Ward) Clustering

Ward linkage merges, at each step, the pair of clusters that minimises the increment in total within-cluster variance, favouring compact, approximately spherical groups. As Figure 10 shows, the dendrogram exposes the full merge hierarchy and its largest vertical gap corresponds to a three-cluster cut; the number of clusters is confirmed by maximising the silhouette over the candidate cuts, yielding $k = 3$ in agreement with K-means. The resulting partition is shown in Figure 11.

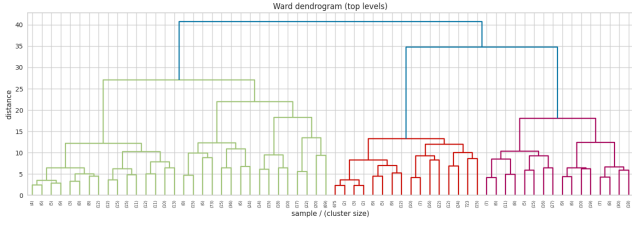


Fig. 10. Truncated Ward dendrogram (top levels). The largest vertical gaps indicate the natural number of clusters.

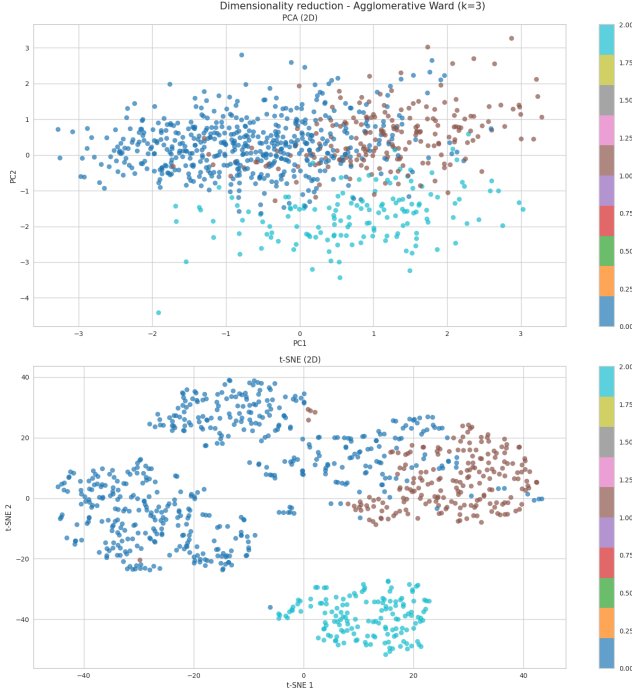


Fig. 11. Agglomerative (Ward) partition in the PCA and t-SNE projections.

D. DBSCAN

DBSCAN requires two parameters: the neighbourhood radius ϵ and the minimum number of points min_samples . A principled estimate of ϵ is obtained from the knee of the sorted 4-nearest-neighbour distance curve (Figure 12), located automatically with the `KneeLocator` heuristic. This initial estimate is refined by a small grid search over ϵ and min_samples , selecting the configuration that maximises the silhouette computed on the non-noise points. Because DBSCAN optimises no global k , it is free to label sparse observations as noise (-1); the resulting partition, with noise points shown separately, appears in Figure 13.

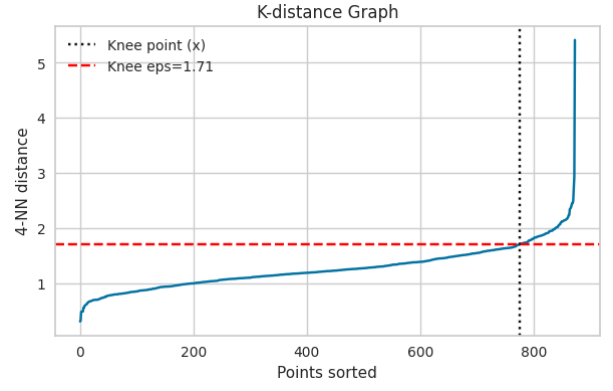


Fig. 12. Sorted 4-NN distance curve. The knee (dashed line) provides the base estimate of the DBSCAN radius ϵ .



Fig. 13. DBSCAN partition in the PCA and t-SNE projections. Noise points (-1) are shown separately from the dense clusters.

VI. EVALUATION AND RESULTS

A. Defining Optimal Clustering

In the absence of ground-truth labels, *optimal* clustering is defined purely through internal validation. The Silhouette Coefficient is adopted as the primary metric: for each point it contrasts the mean intra-cluster distance with the mean nearest-cluster distance, producing a score in $[-1, 1]$ where higher values denote well-separated, cohesive clusters. The Davies–Bouldin index (lower is better), which measures the average ratio of intra-cluster scatter to inter-cluster separation, is used as a corroborating criterion. A partition is preferred when it simultaneously maximises the silhouette and minimises Davies–Bouldin.

TABLE II

INTERNAL VALIDATION OF THE THREE CLUSTERING ALGORITHMS ON THE STANDARDISED, PCA-REDUCED FEATURE SPACE. HIGHER SILHOUETTE AND LOWER DAVIES–BOULDIN INDICATE BETTER STRUCTURE.

Method	k	Silhouette	Davies–Bouldin
K-means++	3	~ 0.16	1.72
Agglomerative (Ward)	3	~ 0.15	1.79
DBSCAN	2	~ 0.40	0.90

The results are summarised in Table II and visualised in Figure 14. DBSCAN produces the cleanest and most stable partition, achieving both the highest silhouette (≈ 0.40) and the lowest Davies–Bouldin (0.90) by isolating the dense core of the data and relegating diffuse observations to the noise label. K-means++ and Ward converge on a common three-cluster solution with more modest but mutually consistent scores (silhouette ≈ 0.15 – 0.16 , Davies–Bouldin ≈ 1.7 – 1.8), reflecting the fact that mixed-type standardised clinical data rarely form widely separated spherical clouds.

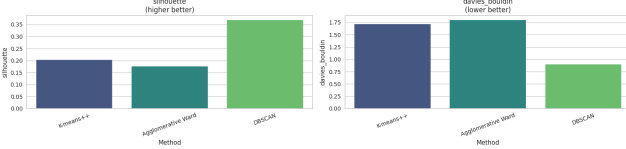


Fig. 14. Comparison of silhouette (higher is better) and Davies–Bouldin (lower is better) across the three algorithms.

B. The Effect of Scaling on the Distance Metric

To make the effect of feature scaling explicit, K-means was applied to three variants of the data: the raw unscaled matrix, the same matrix after standardisation, and the full production pipeline (one-hot encoding, standardisation, and PCA at 90%). As reported in Table III and illustrated in Figure 15, the unscaled configuration yields a deceptively high silhouette (≈ 0.5 – 0.7), which collapses to ≈ 0.16 once the features are standardised. Clustering on the raw `chol` column *alone* reproduces most of the inflated unscaled silhouette, confirming that the apparent quality was an artefact of a single dominant, unnormalised variable rather than genuine multivariate structure. The standardised silhouettes are therefore the honest measure of clustering quality on this dataset.

TABLE III

EFFECT OF FEATURE SCALING ON THE K-MEANS SILHOUETTE. THE INFLATED UNSCALED SCORE REFLECTS A UNIVARIATE SPLIT ON CHOLESTEROL AND COLLAPSES ONCE FEATURE PARITY IS RESTORED.

Pipeline	Sil. $k=2$	Sil. $k=3$
Unscaled	~ 0.70	~ 0.50
Same data, standardised	~ 0.20	~ 0.16
Full pipeline (one-hot + scaled + PCA-90%)	~ 0.17	~ 0.16

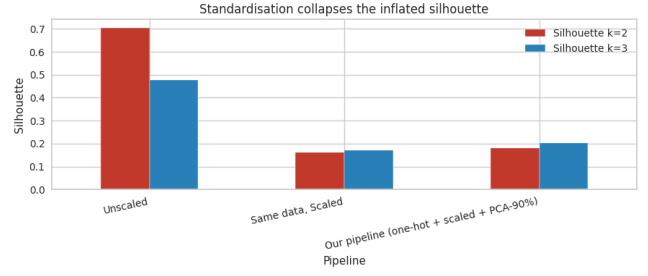


Fig. 15. Standardisation collapses the inflated silhouette obtained on the raw matrix, exposing the true difficulty of the clustering problem.

VII. ALGORITHM ANALYSIS: CONVERGENCE, LOCAL MINIMA, AND STABILITY

A. K-means++

K-means minimises a non-convex within-cluster SSE objective:

$$\text{SSE} = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2, \quad (1)$$

where K is the total number of clusters, C_k is the set of data points assigned to the k -th cluster, and μ_k is the centroid (mean vector) of cluster C_k . Given this objective, the algorithm is guaranteed to converge only to a *local* minimum; unfortunate seed placement can yield poor partitions. Two mitigations are applied: the `k-means++` seeding scheme spreads initial centroids apart to reduce the probability of a bad start, and $n_{\text{init}} = 10$ independent restarts are performed, retaining the solution of lowest SSE. Convergence within each restart is declared when the centroid shift falls below the tolerance 10^{-4} or the iteration cap of 300 is reached.

B. Agglomerative (Ward)

Agglomerative clustering is *deterministic*: the merge sequence is a fixed function of the distance matrix, so there is no random initialisation and no local-minima concern—repeated runs produce identical output. Its limitation is greediness: an early merge is never reversed, so the final partition can be globally sub-optimal even though it is perfectly reproducible.

C. DBSCAN

DBSCAN does not optimise a global objective and thus cannot become trapped in a local minimum. It is deterministic up to the assignment of *border* points, whose cluster membership can depend on processing order. Its principal sensitivity is to ϵ ; this is controlled by anchoring the radius to the k -distance knee rather than choosing it arbitrarily.

D. Handling of Outliers and Noise

The three algorithms treat anomalous observations very differently. K-means and Agglomerative clustering are *not* noise-aware: every point must be assigned to some cluster, so outliers are absorbed into the nearest group and can distort centroids and merge decisions. DBSCAN is intrinsically robust, explicitly labelling low-density points as noise (-1) and thereby preventing them from contaminating the dense clusters. To

protect the two non-robust methods, an Isolation Forest pre-filter (Section III-C) removes the 5% most anomalous records before clustering, so that all three algorithms operate on a comparably clean feature space.

E. Reproducibility

Every stochastic component—K-means seeding, the t-SNE embedding, the randomised PCA solver, and the Isolation Forest—is seeded with a fixed `RANDOM_STATE` of 42. Consequently the entire pipeline is reproducible: re-execution yields identical clusters, metrics, and figures.

VIII. BAYESIAN NETWORK: PROBABILISTIC STRUCTURE DISCOVERY

A. Discretisation and Structure Learning

To complement the geometric clustering with a probabilistic view of the data, a Bayesian Network is learned over the clinical variables augmented with the discovered cluster label. Because Bayesian Networks require discrete variables for conditional-probability estimation, the continuous features are binned into five quantile bins with `KBinsDiscretizer`, and categorical variables are label encoded. The best K-means partition is appended as an additional `Cluster` node.

The directed acyclic graph (DAG) is discovered by Hill-Climb Search [4], a local search that iteratively applies the single edge addition, deletion, or reversal that most improves a scoring function, subject to a maximum in-degree of three. The search is scored with the discrete Bayesian Information Criterion (BIC).

B. Information Criteria and Complexity Penalisation

The BIC score balances goodness of fit against model complexity:

$$\text{BIC} = -2\ell(\hat{\theta}) + k \log n, \quad (2)$$

where $\ell(\hat{\theta})$ is the maximised log-likelihood, k the number of free parameters, and n the sample size. The $k \log n$ term penalises each additional edge—and hence each enlargement of a conditional probability table—discouraging spurious dependencies and yielding a sparse, interpretable structure. For comparison, the Akaike Information Criterion,

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k, \quad (3)$$

of the same learned graph was also computed. As summarised in Table IV, because the BIC penalty $k \log n$ is strictly larger than the AIC penalty $2k$ for any $n > 7$, BIC favours the sparser DAG, which is preferable in a clinical setting where interpretability and resistance to overfitting are paramount.

C. Topology and Inference

The learned graph is analysed with betweenness centrality, which, as Figure 16 shows, identifies the `Cluster` node as the most central variable: it directly parents many clinical findings (`cp`, `chol`, `exang`, `thalch`, `restecg`), acting as a latent risk stratifier off which the observable variables hang. Conditional probability tables are estimated with a Bayesian (BDeu) estimator using a weak prior (equivalent sample size

TABLE IV
INFORMATION-CRITERION COMPARISON OF THE HILL-CLIMB LEARNED STRUCTURE. THE STRICTER BIC PENALTY YIELDS THE SPARSER, MORE INTERPRETABLE GRAPH.

Criterion	Penalty term
BIC (selection score)	$k \log n$ (stricter $\Rightarrow$ sparser DAG)
AIC (reported)	$2k$ (more permissive)

= 5) so that rare parent configurations are smoothed rather than assigned degenerate 0/1 probabilities.

Exact inference is performed with Variable Elimination. Querying $P(\text{Cluster})$ and its conditionals—for example conditioning on the lowest versus highest cholesterol bin—reveals that the clusters shift the clinical findings substantially away from the population marginal. As quantified in Figure 17, one cluster concentrates asymptomatic chest pain and exercise-induced angina with reduced maximal heart rate (a high-burden “silent ischaemia” phenotype), while its mirror image exhibits typical/atypical angina, low angina probability, and preserved heart rate (a lower-risk phenotype). These consistent, interpretable shifts confirm that the discovered partition behaves as distinct cardiac phenotypes rather than noise.

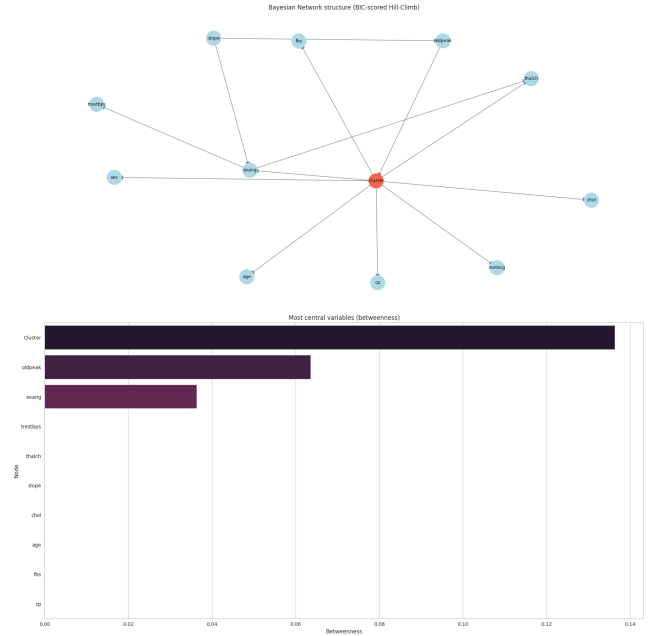


Fig. 16. Learned Bayesian Network (BIC-scored Hill-Climb search) and the betweenness centrality of its nodes. The `Cluster` node (highlighted) is the most central variable.

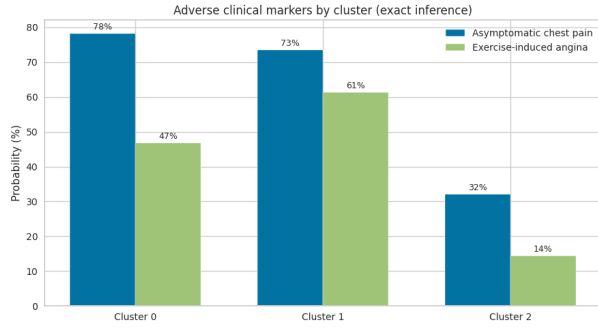


Fig. 17. Probability of adverse clinical markers (asymptomatic chest pain and exercise-induced angina) by cluster, obtained by exact inference. The separation across clusters confirms distinct phenotypes.

IX. COMPUTATIONAL COMPLEXITY AND SCALABILITY

The suitability of each technique for a real-time environment with millions of data points depends critically on its asymptotic cost, summarised in Table V for n samples, d dimensions, k clusters, and t iterations.

TABLE V
TIME AND MEMORY COMPLEXITY OF THE THREE TECHNIQUES.

Algorithm	Time	Memory
K-means++	$O(n k d t)$	$O(n d)$
Agglomerative (Ward)	$O(n^2 d + n^2 \log n)$	$O(n^2)$
DBSCAN	$O(n \log n) - O(n^2)$	$O(n)$

As Table V makes clear, K-means++ scales *linearly* in the number of samples, $O(n k d t)$, and admits a mini-batch variant that processes the data in streaming fashion; it is by far the most viable candidate for real-time deployment at the scale of millions of points. Agglomerative Ward clustering, by contrast, requires the full $O(n^2)$ pairwise distance matrix and at least $O(n^2 \log n)$ time, rendering it memory-prohibitive beyond tens of thousands of samples—it is best reserved for offline, exploratory analysis of the merge hierarchy. DBSCAN occupies an intermediate position: with a spatial index such as a KD-tree or ball-tree its expected cost is $O(n \log n)$ in low dimensions, but it degrades toward $O(n^2)$ as dimensionality grows and the index loses effectiveness. For a genuinely real-time, large-scale system, (mini-batch) K-means is therefore the recommended technique, with PCA applied upstream to keep d small and preserve the efficiency of the distance computations.

X. CONCLUSION

An end-to-end unsupervised pipeline was developed to recover cardiac risk phenotypes from the UCI Heart Disease dataset without any diagnostic label. After removing non-informative and high-missing columns and filtering 5% of outliers with an Isolation Forest, PCA at 90% variance reduced the encoded space from 19 to 10 dimensions, mitigating the curse of dimensionality. Three structurally distinct clustering algorithms were compared under a strict internal validation

regime: K-means++ and Ward converged on a common three-cluster solution, while DBSCAN produced the cleanest, most stable partition (silhouette ≈ 0.40) by explicitly isolating noise. A controlled experiment demonstrated that standardisation is indispensable—the high silhouette obtained without it was an artefact of an unnormalised cholesterol axis. Finally, a BIC-scored Bayesian Network recovered an interpretable dependency structure in which the discovered cluster label emerged as a central latent risk stratifier, and exact inference confirmed that the clusters correspond to distinct, clinically meaningful cardiac phenotypes. For deployment at the scale of millions of streaming records, mini-batch K-means, preceded by PCA, offers the most favourable balance of quality and computational scalability.

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